



Launching VPC Resources



Arpita Chowdhury

VPC settings

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Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

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IPv4 CIDR block [Info](#)

Determine the starting IP and the size of your VPC using CIDR notation.

65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)

Number of Availability Zones (AZs) [Info](#)

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

Preview

VPC [Hide details](#)

Your AWS virtual network

nextwork-vpc

10.0.0.0/16

No IPv6

Subnets (6)

Subnets within this VPC

us-east-1a

network-subnet-public1-us-east-1a

10.0.0.0/20

No IPv6

network-subnet-private1-us-east-1a

10.0.128.0/20

No IPv6

network-subnet-private3-us-east-1a

10.0.192.0/20

No IPv6

us-east-1b

network-subnet-public2-us-east-1b

10.0.16.0/20

No IPv6

network-subnet-private2-us-east-1b

10.0.144.0/20

No IPv6

network-subnet-private4-us-east-1b

Route tables (5)

Route network traffic to resources

network-rtb-public

0.0.0.0/0 routes to network-igw

network-rtb-private1-us-east-1

Prefix list through network-vpc-c3

network-rtb-private2-us-east-1

Prefix list through network-vpc-c3

network-rtb-private3-us-east-1

Prefix list through network-vpc-c3

network-rtb-private4-us-east-1

Prefix list through network-vpc-c3



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a Virtual Private Cloud that lets you create an isolated, customizable network environment within AWS. It is useful because it gives you full control over your networking including IP address ranges, subnets, route tables, gateways, and security groups — so you can securely run resources (like EC2 instances, databases, and applications) in the cloud.

How I used Amazon VPC in this project

I used Amazon VPC to design a secure network for my project by creating both public and private subnets, attaching an internet gateway for public access, and configuring route tables and security groups to control traffic. This allowed me to place resources like web servers in the public subnet and databases in the private subnet, ensuring both accessibility and security.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was using VPC and more were all other resources like route table, security group, internet gateway and network ACL are all created simultaneously and not needed to be created separately.



This project took me...

This project took me 2 hours.



Setting Up Direct VM Access

Directly accessing a virtual machine means connecting to your EC2 instance itself (for example, via SSH for Linux or RDP for Windows) using its IP address and key pair, without going through a load balancer, bastion host, or other intermediate service.

SSH is a key method for directly accessing a VM

SSH traffic means the secure, encrypted communication exchanged between a client and a server using the SSH (Secure Shell) protocol, typically over TCP port 22.

To enable direct access, I set up key pairs

A key pair are used for secure login to EC2 instances without needing a password.

A private key's file format means the type of file extension used to store the key material (such as .pem or .ppk). My private key's file format was .pem.



Launching a public server

I had to change my EC2 instance's networking settings by selecting the NextWork VPC for VPC, NextWork public subnet for subnet, existing security group for security group and NextWork security group for common security group.

Instance summary for i-010ada20a285f652a (NextWork Public Server) Info		
Updated less than a minute ago		
Instance ID i-010ada20a285f652a	Public IPv4 address 35.153.162.121 open address	Private IPv4 addresses 10.0.0.61
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-0-61.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-0-61.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 35.153.162.121 [Public IP]	VPC ID vpc-0b4112e76cd834299 (NextWork)	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-09505029f94079b78 (NextWork Public Subnet)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:006834263464:instance/i-010ada20a285f652a	
Operator -		

Launching a private server

My private server has its own dedicated security group because it requires stricter access controls, allowing only specific internal traffic (for example, from the public server or a load balancer) while blocking direct access from the internet.

My private server's security group's source is another security group (the public server's SG), which means only instances that belong to that source security group are allowed to initiate SSH connections into my private server.

Security group name - *required*

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-:/#,@!+=&:~|!*'()*[]

Description - *required* | [Info](#)

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, sg-013b8ae50c887c59d)

Type | [Info](#)

Protocol | [Info](#)

Port range | [Info](#)

Source type | [Info](#)

Source | [Info](#)

[✕](#)

Description - *optional* | [Info](#)

[Add security group rule](#)

[► Advanced network configuration](#)

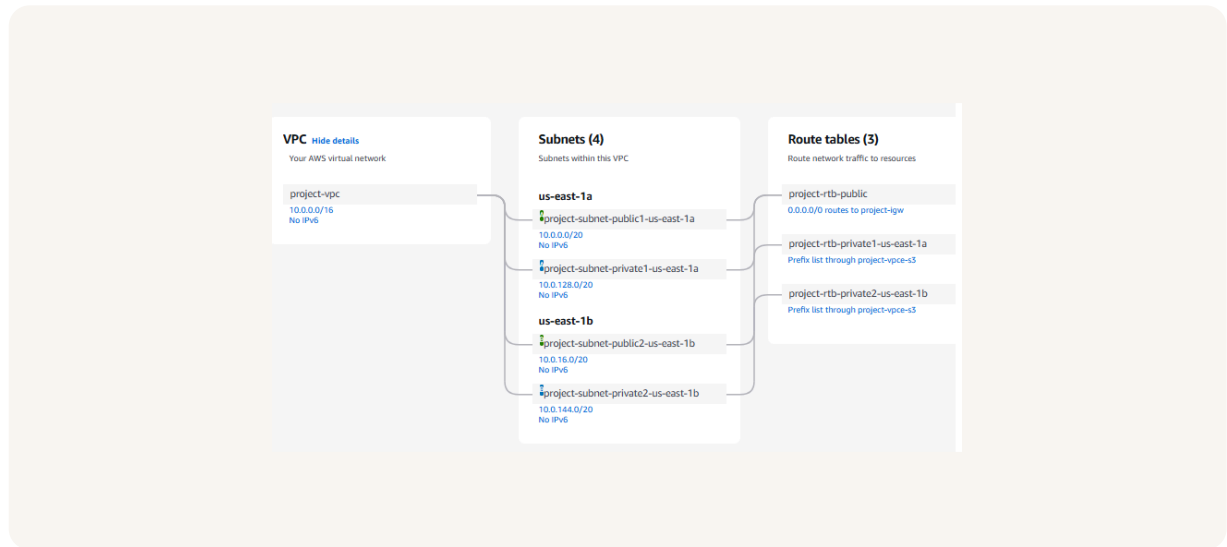


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I used VPC and more in the VPC console wizard, which automatically created a VPC with public and private subnets, route tables, and optional NAT gateways across two Availability Zones. This setup follows AWS best practices for high availability and redundancy, making the network production-ready from the start.

A VPC resource map is a visual diagram that shows all the networking resources inside a Virtual Private Cloud (VPC) such as subnets, route tables, security groups, gateways, and instances and how they are connected to each other.

My new VPC has a CIDR block of 10.0.0.0/16. It is possible for my new VPC to have the same IPv4 CIDR block as my existing VPC because VPCs are isolated virtual networks, and their IP ranges do not overlap unless I explicitly connect them (for example, through VPC peering or a Transit Gateway).





Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: 0 and 2. This was because AWS' best practice advice is built into the VPC wizard. If I select two Availability Zones, the wizard automatically creates a public subnet in each one. This ensures redundancy (having backups in different places) and high availability (keeping resources accessible even if one AZ goes down). Just one public subnet would not provide this reliability, so the option isn't offered. The wizard also limits me to two public subnets for simplicity, but I can always add more manually later if needed (up to 200 in a VPC).

The setup page also offered to create NAT gateways, which are managed AWS services that let private subnets access the internet securely without exposing those resources to inbound internet traffic.



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nextwork

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